Kurt Matthew Nudo

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WORK EXPERIENCE

International Business Machines (IBM) **Feb. 2026 – Present**
Experienced Full Stack Developer for Banking and Financial Markets, Japanese Bilingual *Quezon City, NCR*

- Deliver backend modules for a large-scale Japanese banking migration, using Java 17, Jakarta EE, Maven, MyBatis, Oracle, JUnit, and Mockito.
- Draft initial module implementations from reviewed Japanese design documents, then integrate, inspect, debug, and refine them by hand before release.
- Translate Japanese technical specifications into production-ready business logic, contributing to 20+ migrated modules while maintaining financial and transaction integrity.
- Write and maintain 50+ unit tests by hand, covering test data, database setup, exception handling, and verification of business logic and database interactions.
- Diagnose and resolve 20+ defects and edge cases through code-level analysis and database verification, reducing integration and release issues.
- Apply DevOps practices across automated testing, continuous integration, deployment, and release validation.

Advanced World Solutions **Jul. 2024 – Feb. 2026**
Full Stack Developer, Systems Migration Engineer (Legacy C# System → React + TypeScript, Node.js) *Makati, NCR*

- Worked on a long-term migration of a legacy C# desktop system for a Japanese client to a modern web architecture using TypeScript, React, and Node.js.
- Achieved >95% feature parity while modernising the codebase with a modular architecture and asynchronous data handling.
- Collaborated directly with Japanese engineers to clarify specifications, resolve ambiguous requirements, and keep the system's language accurate.
- Migrated 20+ production modules from the desktop system to the web stack for a Japanese enterprise client within the first year.
- Integrated Oracle databases through REST APIs for data persistence, query optimisation, and transaction management.
- Resolved 40+ verified defects across UI workflows, API integrations, and localised Japanese content.
- Developed unit and integration tests with Jest for every migrated module, with ≥90% code coverage as a release criterion.

PROJECTS

Typhoon Replay · typhoon.krtmtthw.dev *2026*

- Hour-by-hour replays of typhoons that hit the Philippines or Japan since 1940 (ERA5 reanalysis beside the official best track), in English and Japanese. React and TypeScript, a Java 17 Spring Boot API on PostGIS, and a Python pipeline in GitHub Actions that rebuilds and archives the data.

Lights · lights.krtmtthw.dev *2026*

- Poverty estimates for every Philippine city and municipality, 2012–2025: PSA official figures plus labelled model estimates with calibrated uncertainty, building on the publication below. Earth Engine satellite data, LightGBM, a Rust core compiled to WebAssembly, and a MapLibre map.

Akari design system *2026*

- React and TypeScript design system (design tokens, shadcn/ui components, light, dark, and high-contrast themes) shared by every app above.

PUBLISHED LITERATURE

Temporal Analysis of Nighttime Lights and Vegetation Indices for Poverty Forecasting: A Recurrent Neural Network Perspective
Springer, Lecture Notes in Networks and Systems, 2026, pp. 451–461 · co-author

- Study on remote sensing methods for poverty forecasting, using vegetation indices and nighttime light intensity as key variables. Extracted data from Google Earth Engine and government databases, designed the model, and led the analysis with recurrent neural networks.
- Technologies: Python, TensorFlow, JavaScript for Earth Engine.

EDUCATION AND CERTIFICATION

Saint Louis University **Jul. 2024**
Bachelor of Science in Computer Science *Baguio, Benguet*

- Graduated Summa Cum Laude with a general average of 93.

Japanese-Language Proficiency Test (JLPT) N4 *studying for N3*